

Cont...

Introduction to Histology

- Objectives
- ✓ Define what is histology
- ✓ Describe the cell structure
- ✓ Explain what is tissue
- ✓ Describe the procedures of tissue preparation

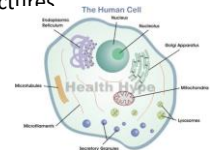
Cont....

- Histology is the study of the **tissues** of the body
- **how** these tissues are **arranged** to constitute organs
- Tissues have two interacting components:
 - ✓ cells
 - ✓ extra-cellular matrix(ECM)



The cell

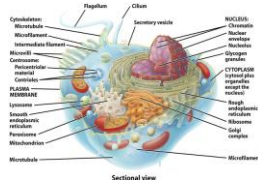
- The **smallest structural and functional units**.
- capable of carrying out all the functions of living things
- self-contained and fully operational living entity
- **Eukaryotic:** **distinct** membrane-limited nuclei surrounded by cytoplasm containing various **membrane-limited** organelles
- **Prokaryotic:** have a cell wall around the plasmalemma and **lack** other membranous structures



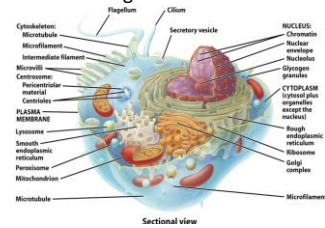
Cont...

Cont...

- The cell contains various structural components to allow it to maintain life which are known as **organelles**
- All the organelles are suspended within a **gelatinous** matrix, the **cytoplasm**,
- ✓ which is **contained** within the **cell membrane**
- One of the few cells in the human body that **lacks** almost all organelles are the **red blood cells**

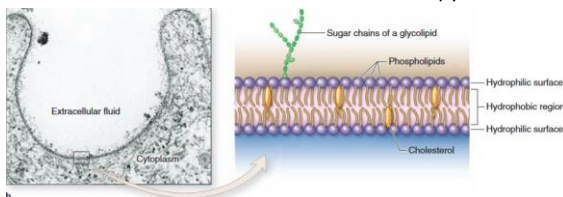


- Animal cells are eukaryotic cells.
- Each cell carries out basic life processes that allow the body to survive.
- All human cell are **microscopic in size, shape and function.**
- The diameter range from 7.5 micrometer to 150 μ m.



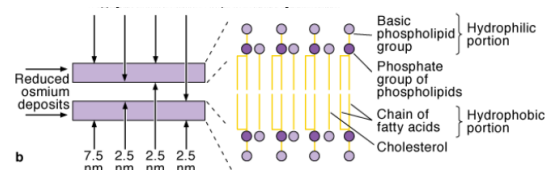
Cell membrane

- is thin, dynamic membrane that encloses the cell
- actively participates in many **physiologic** and **biochemical** activities essential to cell function and survival
- Visible with transmission electron microscopy



Cont...

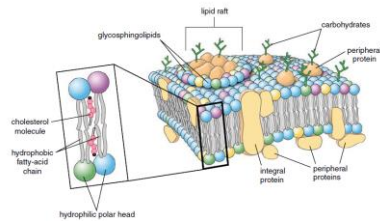
- composed of **phospholipids**, **cholesterol**, **proteins**, and chains of **oligosaccharides**
- The lipid molecules form a lipid bilayer with an **amphipathic** character
- range from 7.5 to 10 nm in thickness



Source: Mescher AL : *Jurqueira's Basic Histology: Text and Atlas*

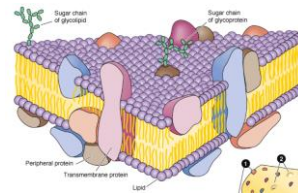
Cont...

- The orientation of the phospholipids form the **bilayer** of **biological** membranes
- **hydrophilic** polar heads, toward each surface of the **membrane**
- **hydrophobic** heads, **away from water**



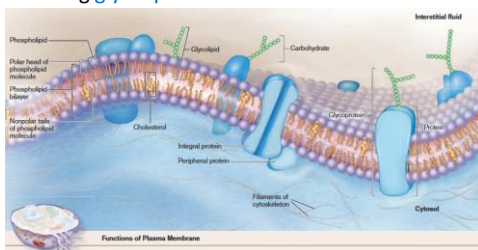
Cont...

- Membrane proteins are **Integral** proteins and **peripheral** proteins
- integrins are linked to both cytoplasmic **cytoskeletal filaments** and **extracellular** matrix component
- **integral** membrane proteins are **embedded** within the lipid bilayer or **pass** through the lipid bilayer completely



Cont...

- **peripheral membrane** proteins are not embedded within the **lipid bilayer**
- Associate with the plasma lipids of the bilayer, thereby forming **glycolipids**

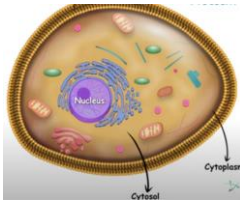


Cont...

- **Functions:**
 - as a **selective barrier** that regulates the passage of certain materials into and out of the cell
 - the **transport** of specific molecules
 - keep constant the **ion** content of the cytoplasm
 - carry out a number of specific **recognition** and **regulatory** functions
 - Play an important role in the **interactions** of the cell with its environment

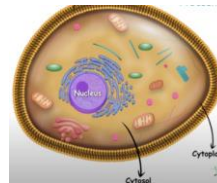
Cytoplasm

- Is a **gel-like** matrix of water, enzymes, nutrients, wastes, and gases
- contains cell structures (**organelles**)
- Also contains **inclusions** which are generally deposits of **carbohydrates, lipids, or pigments**.



Cont...

- The shape and motility of eukaryotic cells are determined by components of the **cytoskeleton**
- Fluid around the organelles called **cytosol**
- Most of the cells **metabolic reactions** occur in the cytoplasm

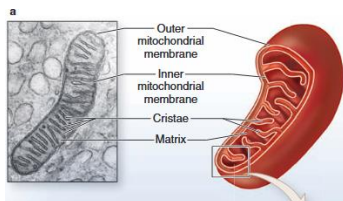


Cytoplasmic organelles

- Can be membranous or non-membranous protein complexes.

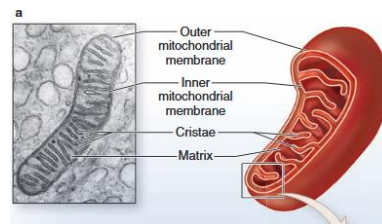
Mitochondria

- are membrane-enclosed organelles
- have **two** membranes



Cont...

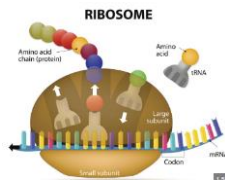
- The inner membrane is folded to form **cristae**
- Enzymes specialized for **aerobic** respiration and production **ATP**
- cells with a high-energy metabolism have abundant **mitochondria**



Cont...

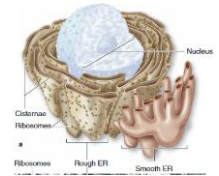
Ribosomes

- Ribosomes are composed of **rRNA** and **proteins**
- all ribosomes have *large* and *small* ribosomal subunits
- Bound to **mRNA**, act to catalyze the process of protein **translation**
- assemble the protein chain from **amino acids** **ferried** in by **tRNA**



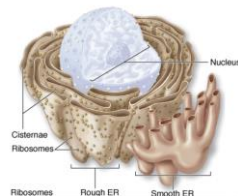
Endoplasmic Reticulum

- Network of **interconnected** parallel membranes, that is continuous with the **nuclear membrane**
- continuous membrane encloses a space called **cisterna**
- Two types:
 - Rough Endoplasmic Reticulum
 - and Smooth Endoplasmic Reticulum
- ✓ **continuous** with one another.



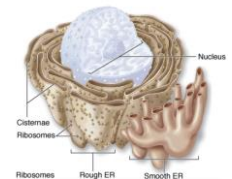
Rough Endoplasmic Reticulum

- Its name indicates the presence of **polyribosomes** on the cytosolic surface of its membrane.
- prominent in cells specialized for **protein** secretion such as:
 - Pancreatic acinar cells
 - Fibroblasts
 - sequestration of Ca^{++} ions.



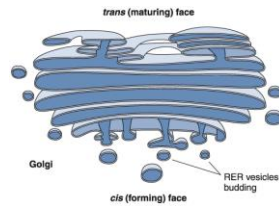
Smooth Endoplasmic Reticulum (SER)

- Regions of ER that **lack** bound polyribosomes.
- In cells that synthesize **steroid hormones** it occupies a large portion of the cytoplasm
- abundant in **liver** cells
- major role is synthesis of various **phospholipid** molecules
- ✓ Have three major Functions:
 - Lipid biosynthesis
 - Detoxification
 - Sequestration of Ca^{++} ions



Golgi Apparatus

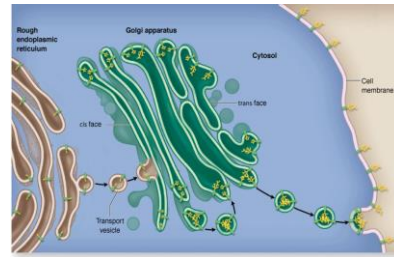
- completes **posttranslational modification** of proteins synthesized in the RER
- Has
 - ✓ Cisterna
 - ✓ Lumen
 - ✓ Transport vesicle, incoming or outgoing
 - ✓ Cis face
 - ✓ Trans face



Cont...

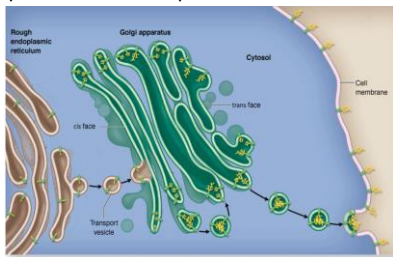
- Functions:

1. **Modification:** modifies new **proteins** destined for **lysosomes**, **secretion**, and **plasma** membrane



Cont...

2. **Packaging:** **packages enzymes** for lysosomes and proteins for secretion
3. **Sorting:** sorts **all materials** for lysosomes, secretion, and incorporation in to the plasma M.



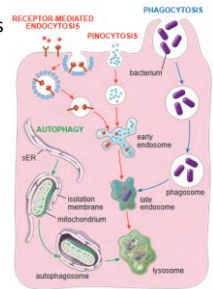
Lysosomes

- are sites of **intracellular digestion** and **turnover** of cellular components.
- abundant in cells with great **phagocytic** activity (eg, macrophages, neutrophils).
- contain about 40 different **hydrolytic** enzymes.
- have a unique membrane that is resistant to the hydrolytic digestion occurring in their lumen

Cont...

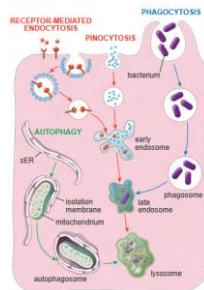
- During lysosomal digestion of macromolecules, released nutrients diffuse into the cytosol through the lysosomal membrane.
- Indigestible materials are retained within the vacuoles and called **residual bodies**
- In some long-lived cells residual bodies can accumulate and are referred to as **lipofuscin granules**.

- Pathways of delivery of materials for digestion in lysosomes
- ✓ Most of the small extracellular particles are internalized by both receptor-mediated endocytosis and pinocytosis
- These two endocytic pathways are labeled with red arrows
- ✓ Large extracellular particles such as bacteria and cellular debris are delivered for cellular digestion via the phagocytotic pathway (blue arrows).

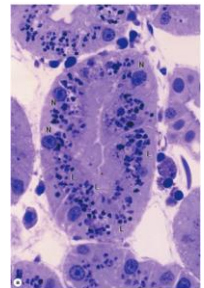


Cont....

- ✓ The cell also uses lysosomes to digest its own organelles and other intracellular proteins via the autophagic pathway (green arrows).
- Intracellular particles are isolated from the cytoplasmic matrix by the isolation membrane of the sER,
- transported to lysosomes, and subsequently degraded.



- lysosomes are **not** well shown on H&E-stained cells but can be visualized by light microscopy after staining with **toluidine blue**
- a) Cells in a kidney tubule show numerous purple lysosomes
- L= in the cytoplasmic area b/n the basally located nuclei = N and apical ends of the cells at the center of the tubule
- using **endocytosis**, these cells actively **take up** small **proteins** in the lumen of the tubule, **degrade** the proteins in lysosomes,
- ✓ and then release the resulting **amino acids** for reuse. X300



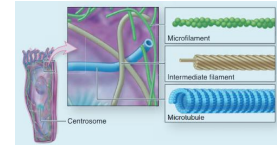
Cont....

Peroxisomes (Microbodies)

- membrane-limited organelles approximately 0.5 micro m in diameter
- contain enzymes that use O_2 to remove hydrogen atoms from substrates
- Oxidize specific organic substrates
- produces hydrogen peroxide (H_2O_2)
- H_2O_2 immediately broken down by *catalase* to water and O_2 .
- also contain enzymes involved in lipid metabolism

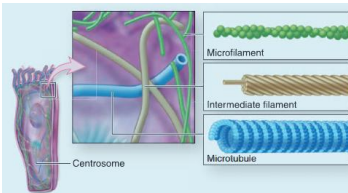
Cytoskeleton

- Nonmembranous organelles
- is a complex network of:
 - microtubules
 - microfilaments
 - intermediate filaments



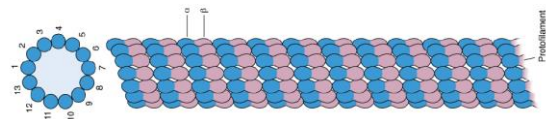
Cont...

- These protein structures:
 - determine the shape of cells
 - play an important role in the movements of organelles and cytoplasmic vesicles,
 - allow the movement of entire cells.



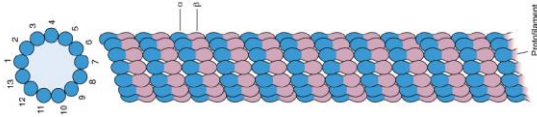
Microtubules

- fine tubular structures within the cytoplasmic matrix.
- also found in cytoplasmic processes called cilia and flagella.
- have an outer diameter of 24 nm, with a dense wall 5 nm thick and a hollow lumen.
- Occasionally, two or more microtubules are linked by protein arms.



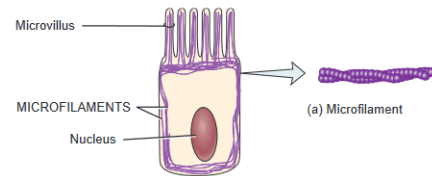
Cont...

- Complex microtubule networks participate in the **intracellular** transport of organelles and vesicles.
- elongated polymeric structures composed of



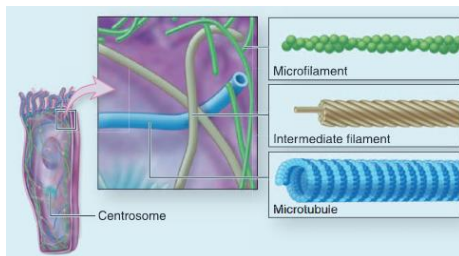
Microfilaments

- are helical two-stranded polymers assembled from **globular actin** subunits
- The filaments are **flexible** structures, with diameters in various cells of **5-9 nm**, depending on associated proteins



Cont...

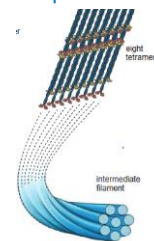
- Contractile activity in cells results primarily from an interaction b/n **actin** and its associated protein, **myosin**



Intermediate Filaments

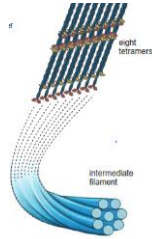
display an average diameter of 10-12 nm, b/n that of actin filaments and microtubules.

- serve to provide **mechanical** strength or **stability** to cells.
- composed of **various protein subunits** in different types of cells.



Cont...

- A large and important class of intermediate filaments is composed of **keratin subunits**.
- Intermediate filament proteins have been organized **chemically** and **genetically** into four major groups:
 - Keratins
 - Vimentin
 - Neurofilaments
 - Lamins

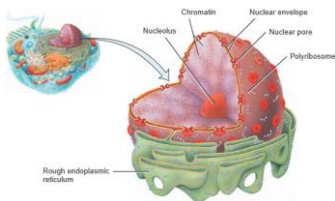


Inclusions

- Cytoplasmic inclusions are composed mainly of **accumulated metabolites** or other substances.
- often they are **transitory** components of the cytoplasm.
- Nonmotile and with little or no metabolic activity, inclusions are **not considered** organelles.
- Important and commonly seen inclusions include:
 - Fat droplets
 - Glycogen granules
 - Lipofuscin granules

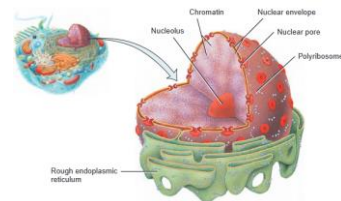
Nucleus

- is the command center of the cell
- is a spherical or oval structure that usually is the most prominent feature of a cell
- Most body cells have a single nucleus
- red blood cells, have no nucleus
- skeletal muscle cells have several nuclei



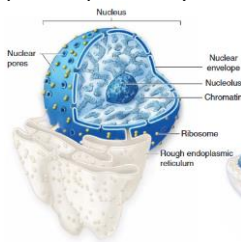
Cont...

- contain the code for all of a cell's enzymes and other proteins
- A double membrane called the nuclear envelope separates the nucleus from the cytoplasm
- Many openings called nuclear pores pierce the nuclear envelope
- one or more spherical bodies called nucleoli are found inside it



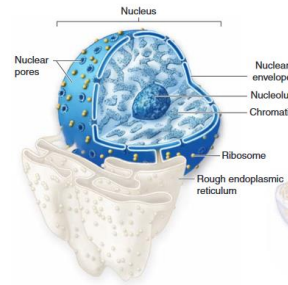
Cont...

- These clusters of protein, DNA, and RNA are the sites of assembly of ribosomes,
- ✓ which exit the nucleus through nuclear pores
- ✓ participate in protein synthesis in the cytoplasm



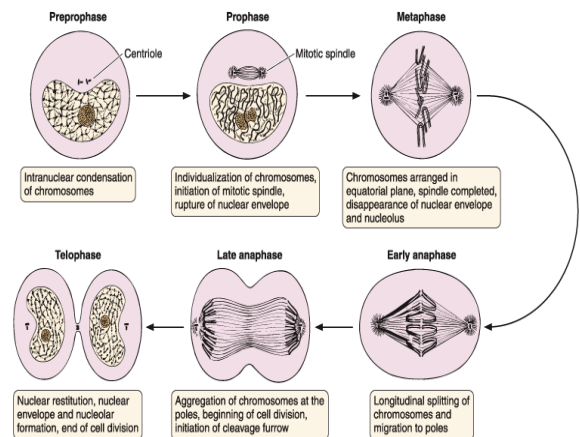
Cont...

- most of the cell's hereditary units, called genes, which control cellular structure and direct most cellular activities are also found within the nucleus

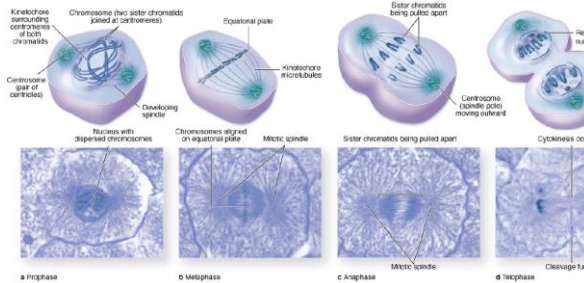


Cell division

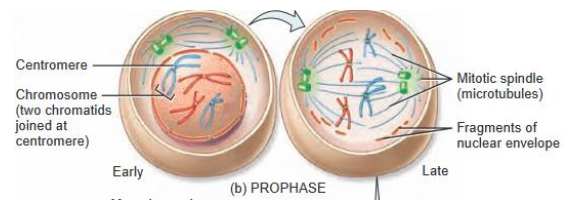
- Cell division, or mitosis can be **observed** with the **light** microscope.
- The parent cell is divided
- each of the daughter cells receives a chromosomal set identical to that of the parent cell.
- The phase b/n two mitoses is called **interphase**.
- subdivided into four phases
 - Prophase
 - Metaphase
 - Anaphase
 - telophase



Cont...

✓ **Prophase:**

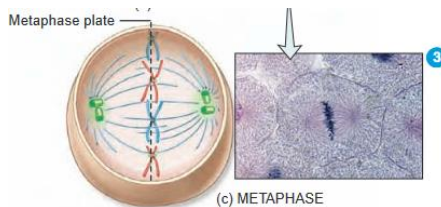
- characterized by the **gradual coiling of nuclear chromatin**, giving rise to several individual rod-shaped bodies.
- The centrosomes with their centrioles separate, and a centrosome migrates to each pole of the cell



Cont...

✓ **Metaphase**

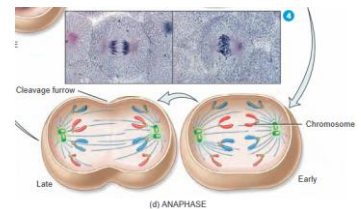
- chromosomes, due to the activity of microtubules, migrate to the **equatorial** plane of the cell.
- where each divides longitudinally to form two chromosomes called **sister chromatids**.



Cont...

✓ **Anaphase**

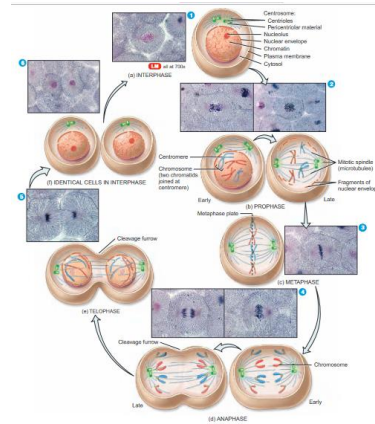
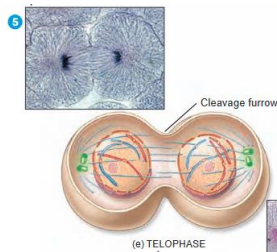
- the sister chromatids separate from each other and migrate toward the **opposite** poles of the cell, pulled by **microtubules**.
- Throughout this process, the centromeres move away from the center, pulling the remainder of the chromosome along.



Cont...

✓ **Telophase**

- characterized by the reappearance of **nuclei** in the **daughter** cells.
- The chromosomes revert to their semidispersed state, and the nucleoli, chromatin, and nuclear **envelope** reappear.

**Tissues**

- Tissues are groups of cells that perform the same function
- the human body is composed of **four** basic types of tissue
 1. Nervous tissue
 2. Epithelial tissue
 3. Connective tissue
 4. Muscle tissue
- These tissues vary in their **composition** and their **function**

Epithelial Tissue

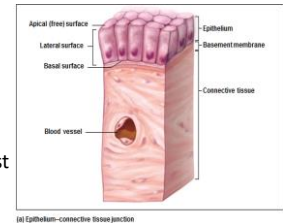
- Epithelial tissue **lines** every body **surface** and **all** body cavities
- Organs** are lined on the **outside** and **inside** by epithelial tissue
- The **majority** of **glands** are derived from epithelial tissue
- ✓ **covering** and **lining** epithelium **and** **glandular** epithelium.
- Epithelial tissue possesses **little** to **no** **extracellular** matrix
- Found **above** a **connective** tissue layer (epi = above)
- Lines the **cavities**, **tubes**, **ducts**, and **blood** vessels inside the body

Cont...

- Functions of epithelial tissue
- Physical protection
- Selective permeability
- Secretions
- Sensations

Characteristics of Epithelial Tissue

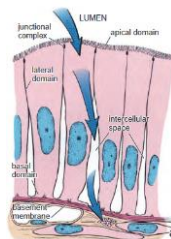
- **Have:**
 - an apical (free) surface
 - lateral surfaces
 - basal surface
- **Innervation**
 - richly innervated
- **Regeneration**
 - replaced as quickly as they are lost
- **Avascular**- Epithelium typically lacks its own blood supply
- cells with **little extracellular material** between them



Cont....

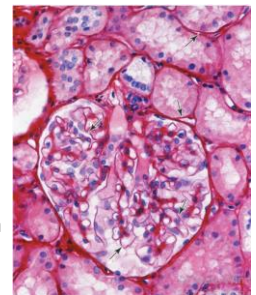
Basement Membrane

- A specialized structure of epithelium
- Found **b/n** the epithelium and underlying connective tissue
- Feltlike sheet of extracellular material
- Provides **physical** support and **anchoring** of epithelial tissue
- Acts as a barrier to **regulate passage** of large molecules b/n epithelium and underlying connective tissue



Cont....

- section of kidney
- well-stained basement membranes = arrows of epithelia forming structures within the large, round renal glomerulus and its surrounding tubules
- in kidney glomeruli the basement membrane, besides having a supporting function, has a highly developed role as a **filter** that is key to renal function. X100. Picrosirius hematoxylin (PSH)



Classifying Epithelia

- bases for classifying covering and lining epithelium are:

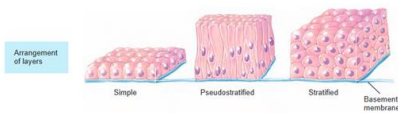
1. Number of cell layers

I. Simple epithelium

- Cells are found in a *single layer* attached to the basement membrane

II. Stratified epithelium

- Cells are found in *2 or more layers* stacked atop each other



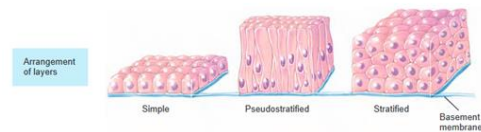
Cont...

III. Pseudostratified epithelium

- a single layer of cells that appears to be multiple layers due to variance in height and location of the nuclei in the cells

IV. Transitional epithelium

- cells are rounded and can slide across one another to allow stretching



2. Cell morphology in the surface layer

1. Squamous

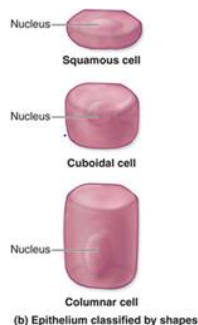
- Flattened, like fish scales

II. Cuboidal

- Cube-shaped, like dice
- Typically the cell's height and width are about equal

III. Columnar

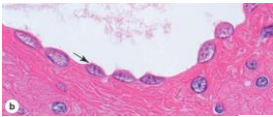
- Column-like
- tall, rectangular or column-shaped cells. Typically taller than they are wide.



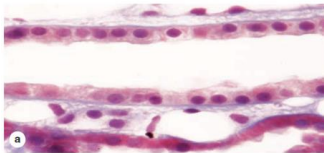
(b) Epithelium classified by shapes

Types of Epithelium

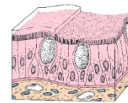
Simple squamous	Vascular system (endothelium) Body cavities (mesothelium) Bowman's capsule (kidney) Respiratory spaces in lung	Exchange, barrier in central nervous system Exchange and lubrication Barrier Exchange
Simple cuboidal	Small ducts of exocrine glands Surface of ovary (germinal epithelium) Kidney tubules Thyroid follicles	Absorption, conduit Barrier Absorption and secretion
Simple columnar	Small intestine and colon Stomach lining and gastric glands Gallbladder	Absorption and secretion Secretion Absorption



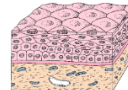
Simple squamous epithelium X600.
H&E.
Simple cuboidal epithelium All X400.
H&E.
Simple columnar epithelium X400.
H&E.



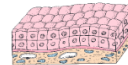
Cont...



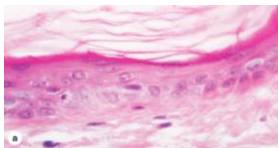
Pseudostratified	Trachea and bronchial tree Ductus deferens Efferent ductules of epididymis	} Secretion, conduit Absorption, conduit
------------------	--	---



Stratified squamous	Epidermis Oral cavity and esophagus Vagina	} Barrier, protection
---------------------	--	-----------------------

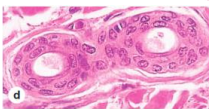
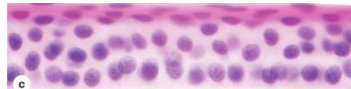


Stratified cuboidal	Sweat gland ducts Large ducts of exocrine glands Anorectal junction	} Barrier, conduit
---------------------	---	--------------------

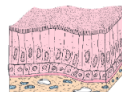


Stratified squamous keratinized epithelia
Stratified squamous nonkeratinized epithelia
Stratified cuboidal or columnar epithelia are fairly rare

All X400; H&E.



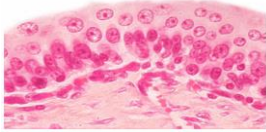
Cont...



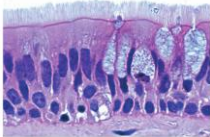
Stratified columnar	Largest ducts of exocrine glands Anorectal junction	} Barrier, conduit
---------------------	--	--------------------



Transitional (urothelium)	Renal calyces Ureters Bladder Urethra	} Barrier, distensible property
---------------------------	--	---------------------------------



transitional epithelium X400, H&E.
pseudostratified columnar epithelium
X400, H&E.



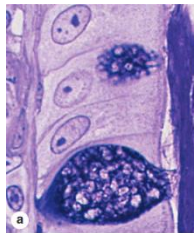
Glandular Epithelium

- Its function is **secretion**
- glandular cells often lie in **clusters** deep to the covering and lining epithelium
- A gland may consist of **one cell** or a group of highly specialized epithelial cells
- secrete substances into *ducts (tubes), onto a surface, or into the blood*

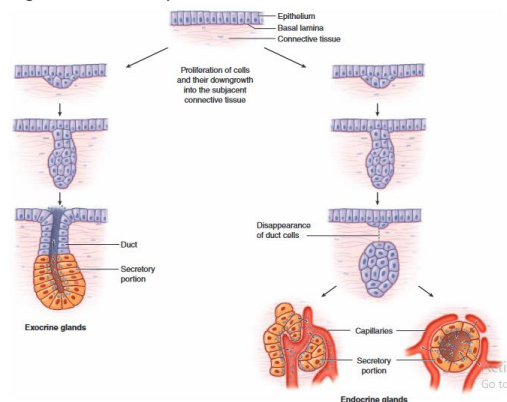
Cont....

Goblet cells: unicellular glands

- The simple columnar epithelium lining the large intestine shows many isolated goblet cells secreting mucus into the lumen = a With a stain for the oligosaccharide components of mucin glycoproteins,
- the cytoplasmic secretory granules of two goblet cells and secreted mucus are *stained purple*. X600.PAS-PT.



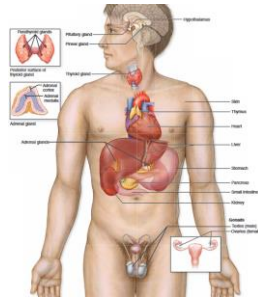
All glands of the body are classified as either endocrine or exocrine



Cont....

Endocrine glands

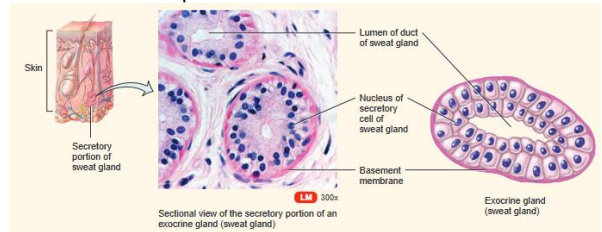
- produce hormones
- hormones enter the interstitial fluid
- ✓ then diffuse into the bloodstream without flowing through a duct
- Examples : pituitary gland, pineal gland, thyroid and parathyroid glands , adrenal glands, pancreas, ovaries, testes, and thymus



Cont...

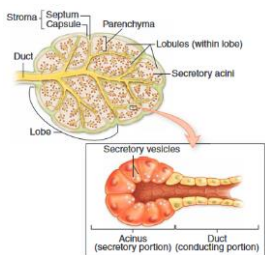
Exocrine glands

- produce substances such as sweat, oil, earwax, saliva, or digestive enzymes
- secrete their products into ducts

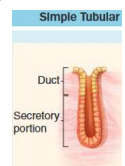


Cont...

- Ducts empty at the surface of covering and lining epithelium such as the skin surface or the lumen
- Examples: Sweat, oil, and earwax glands of the skin; salivary glands and pancreas

**Structural classes of exocrine glands****A. Simple Glands (Ducts Do Not Branch)****1. Simple Tubular**

- **Elongated** secretory Portion
- duct usually **short** or **absent**
- ✓ Examples: Mucous glands of colon; intestinal glands or crypts (of Lieberkühn)

**2. Branched Tubular**

- **Several long** secretory parts
- **joining** to drain into 1 duct
- ✓ glands in the uterus and stomach



3. Coiled Tubular

- Secretory portion is **very long** and **coiled**
- ✓ Sweat glands

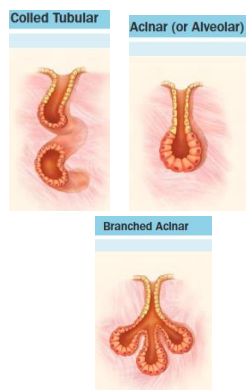
4. Acinar (or Alveolar)

- **rounded, saclike** secretory portion
- ✓ Small mucous glands along the urethra

5. Branched Acinar

- **Multiple** saclike secretory parts entering the same duct
- Sebaceous glands of the skin

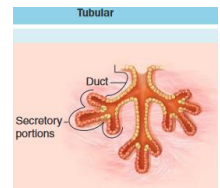
Cont....



B. Compound Glands (Ducts from Several Secretory Units Converge into Larger Ducts)

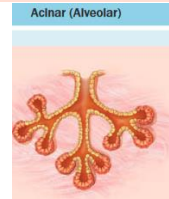
1. Tubular

- Several **elongated, coiled** secretory units
- their **ducts converge** to form larger ducts
- ✓ Examples: Submucosal mucous glands (of brunner) in the duodenum



2. Acinar (Alveolar)

- Several **saclike** secretory units with small ducts converge at a larger duct
- ✓ Exocrine pancreas



Cont....

3. Tubuloacinar

- Ducts of both **tubular** and **acinar** secretory units **converge** at larger ducts
- ✓ Salivary glands

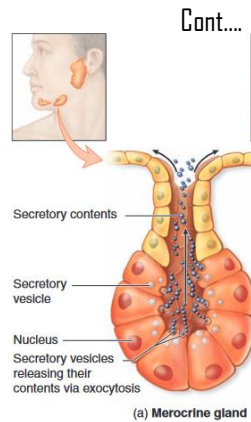


Mechanisms of exocrine gland secretion

- Epithelial cells in **multicellular** glands have three basic mechanisms for releasing their product
- These basic types of secretion are used by cells of exocrine glands, depending on what **substance** is being secreted
- cells involved in each type of secretion are easily recognized histologically

A. Merocrine secretion

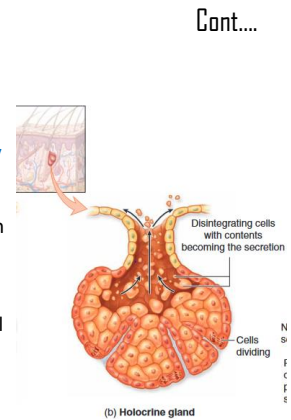
- is the most **common** method of **protein secretion**
- involves typical **exocytosis** of proteins or glycoproteins from membrane-bound vesicles
- ✓ **Most** exocrine glands are merocrine



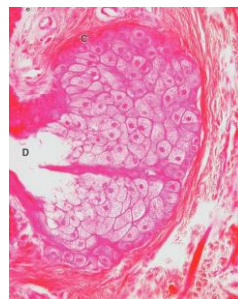
B. Holocrine secretion

- cells accumulate product as they mature
- secretion is produced by the **disintegration** of the **secretory cells themselves**
- as they complete their **terminal differentiation**, which involves becoming filled with product
- **complete** cell **disruption** with release of the product and cell debris into the gland's lumen

✓ **Sebaceous glands**

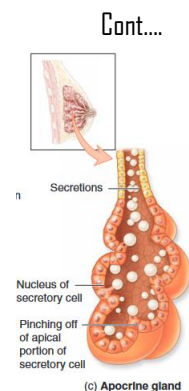


- in holocrine secretion, best seen in the **sebaceous gland** adjacent to hair follicles, entire cells **fill** with a **lipid-rich** product as they differentiate
- Mature (terminally differentiated) cells **separate** and **completely disintegrate**, **releasing** the **lipid** that serves to protect and lubricate adjacent skin and hair
- Sebaceous glands lack **myoepithelial** cells;
- **cell proliferation** inside a dense, inelastic connective tissue capsule continuously forces product into the duct. X200. H&E.



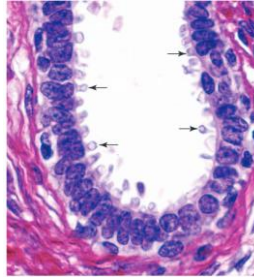
C. Apocrine secretion

- Here product accumulates at the cells' **apical ends**
- **portions** of which are then extruded to release the **product** together with a bit of **cytoplasm** and **plasma membrane**
- secretion involves **loss** of membrane-enclosed apical cytoplasm, usually containing one or more lipid droplets.
- ✓ Apocrine secretion, along with merocrine secretion, is seen in **mammary glands**



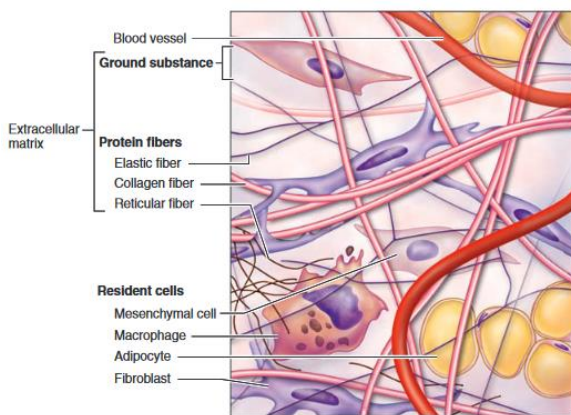
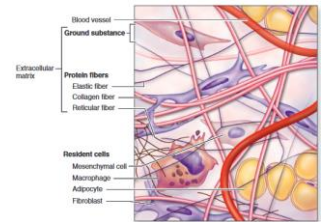
Cont....

- The secretory portions of a mammary gland demonstrate **apocrine** secretion
- characterized by **extrusion** of the **secretion** product along with a **bit** of apical cytoplasm = arrows
- The released portion of cell contains **lipid droplet**= s
- **Merocrine** secretion also occurs from the same and other cells **of the gland**. X400. PSH.



Connective Tissue

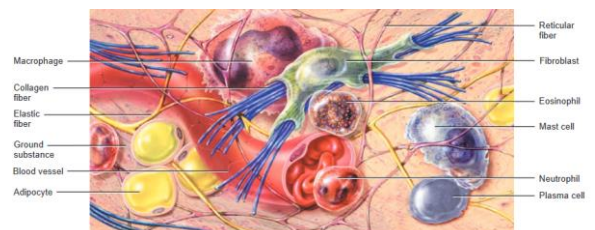
- Most diverse, **abundant** and widely **distributed**.
- consists of two basic elements:
 - ✓ cells
 - ✓ extracellular matrix
 - extracellular matrix consists of **protein fibers** and **ground substance**
 - ground substance is the material b/n the **cells** and the **fibers**



Cont...

✓ Functions:

- Physical protection, Support and structural framework, Binding of structures, Storage, Transport and Immune protection



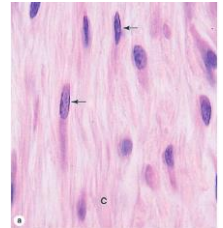
Connective tissue cells

- Can be resident or wandering

1. Fibroblasts

- the principal cell of connective tissue
- *produce* and *maintain* most of the tissue's **extracellular** components
- synthesize and secrete collagen and elastin
- ✓ as well as the GAGs, proteoglycans, and multiadhesive glycoproteins
- that comprise the ground substance
- active and quiescent fibroblasts

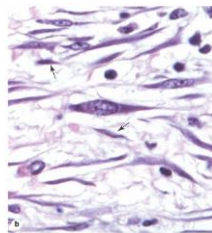
- Fibroblasts typically have **large active** nuclei and **eosinophilic cytoplasm** that tapers off in both directions along the axis of the nucleus,
- ✓ a morphology often referred to as "spindle-shaped."
- Nuclei = arrows are clearly seen,
- ✓ but the eosinophilic cytoplasmic processes resemble the collagen bundles = C that fill the ECM and are difficult to distinguish in H&E-stained sections. X400. H&E.



Cont....

Cont....

- both active and quiescent fibroblasts may sometimes be distinguished, as in this section of dermis
- **active** fibroblasts have **large, euchromatic** nuclei and **basophilic** cytoplasm,
- while **inactive** fibroblasts (or **fibrocytes**) are **smaller** with more **heterochromatic** nuclei = arrows.
- The round, very basophilic round cells are leukocytes. X400. H&E.

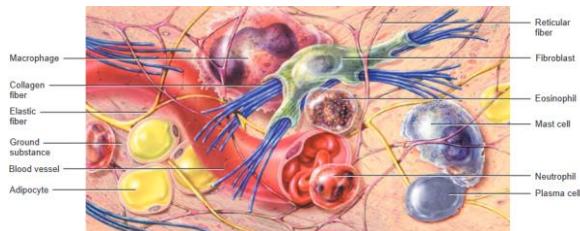


2. Macrophages

- also known as tissue **histiocytes**
- derived from blood cells called **monocytes**
- Monocytes migrate from the bloodstream into the connective tissue,
- ✓ where they differentiate into **macrophages**
- ✓ The transformation involves increases in cell size, increased protein synthesis, and increases in the number of Golgi complexes and lysosomes

Cont...

Cont...



- Macrophages are capable of engulfing **bacteria** and **cellular debris** by **phagocytosis**
- Along with other monocyte derived cells, Macrophages comprise a family of cells called the **mononuclear phagocyte system**

Cont...

3. Mast cells

- develop in bone marrow and differentiate in connective tissue
- Their cytoplasm is filled with basophilic **secretory granules**
- release localized bioactive substances
 - with roles in the local **inflammatory** response, **innate** immunity, and **tissue** repair
- **numerous** near small blood vessels in skin and mesenteries = **perivascular** mast cells
 - and in the tissue that lines digestive and respiratory tracts = **mucosal** mast cells

Cell Type	Major Locations	Main Function
Monocyte	Blood	Precursor of macrophages
Macrophage	Connective tissue, lymphoid organs, lungs, bone marrow, pleural and peritoneal cavities	Production of cytokines, chemotactic factors, and several other molecules that participate in inflammation (defense), antigen processing and presentation
Kupffer cell	Liver (perisinusoidal)	Same as macrophages
Microglial cell	Central nervous system	Same as macrophages
Langerhans cell	Epidermis of skin	Antigen processing and presentation
Dendritic cell	Lymph nodes, spleen	Antigen processing and presentation
Osteoclast (from fusion of several macrophages)	Bone	Localized digestion of bone matrix
Multinuclear giant cell (several fused macrophages)	In connective tissue under various pathological conditions	Segregation and digestion of foreign bodies

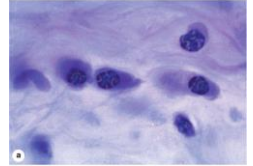
Cont...

4. Plasma cells

- Derived from B-lymphocyte
- synthesis immunoglobulin antibodies,
- proteins that attack or neutralize foreign substances in the body

Cont....

- Plasma cells are large, ovoid cells, with basophilic cytoplasm
- The round nuclei frequently show peripheral clumps
- of heterochromatin, giving the structure a “clock-face” appearance. X640. H&e



Cont...

5. Adipocytes

- also called fat cells or adipose cells
- store triglycerides (fats)
- found below the skin and around organs such as the heart and kidneys
- regulators of the body's overall energy metabolism

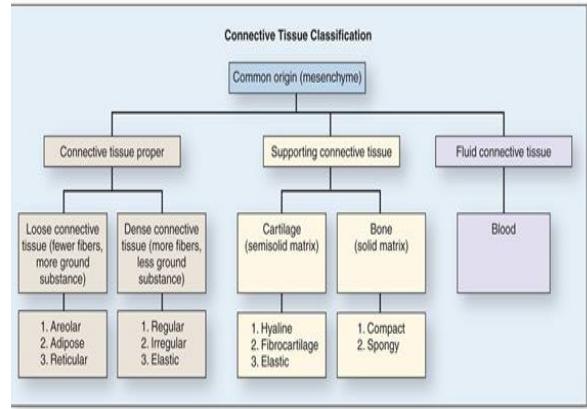
Cont...

6. Leukocytes

- make up a population of wandering cells in connective tissue
- leave the blood by migrating b/n the endothelial cells lining venules to enter connective tissue by a process called diapedesis

Cont...

- CT can be classified into three broad categories:
- ✓ CT **proper**
 - Loose CT
 - Dense CT
- ✓ **Supporting** CT
 - Cartilage
 - Bone
- ✓ **Fluid** CT
 - Comprised of plasma, Erythrocytes, Leukocytes and Platelets



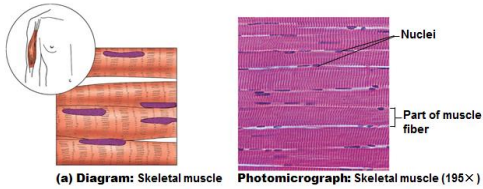
Muscle Tissue

- Comprised of cells called **fibers**.
- When stimulated by the nervous system, fibers shorten or contract.
- The result of contraction is movement (i.e., movement of bones, blood, food, sperm)
- Three types:
 - Skeletal
 - Cardiac
 - Smooth

Cont...

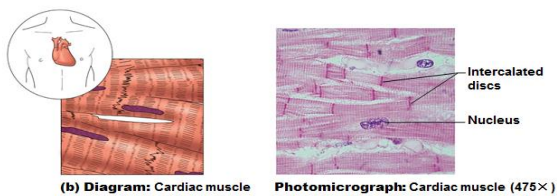
❖ Skeletal muscle

- Attached to bones of skeleton and some skin
- Cells (muscle fibers) are:
 - **cylindrical and long** (some as long as whole muscle)
 - **multinucleated**
 - **striated** (striped internal appearance) and voluntary
- Contraction causes movement of skeleton or skin.



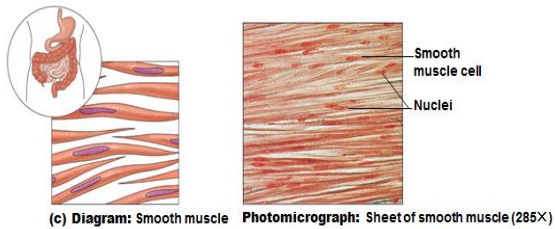
Cardiac Muscle

- Involuntarily controlled
- Found only in the heart
- Pumps blood through blood vessels
- Characteristics of cardiac muscle cells
 - Striations
 - Uninucleate, short, branching cells
 - Intercalated discs contain gap junctions to connect cells together



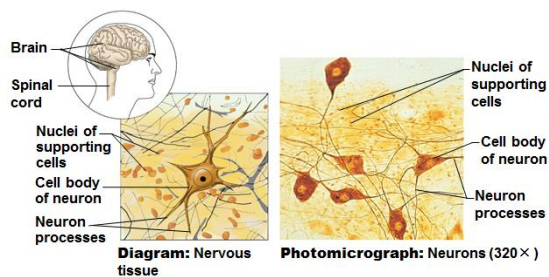
Smooth Muscle

- Found in walls of most internal organs
- stomach, intestines, urinary bladder
- ✓ Cells are:
 - relatively short, wide in the middle, and tapered at the ends (fusiform)
 - involuntary and non-striated
 - Contraction causes movement of food, blood, sperm



Nervous Tissue

- Contains two types of cells:
- ✓ **Neurons:** nerve cells that are capable of initiating and conducting electrical activity throughout the body
- ✓ **Neuroglia:** cells that support the neurons
- Function is communication and control of body functions



Neuron

